Vienna University of Economics and Business

Master Thesis — ML Performance Report

The Agony and the Ecstasy: Machine Learning Model Results & Performance Analysis

Author: Tristan Leiter

Supervisor: Univ.Prof. Dr. Kurt Hornik
Department of Finance, Accounting and Statistics

Date: March 20, 2026

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Chapter 1

Methodology

1.1 Problem Framing and Model Purpose

This thesis trains models to predict **Catastrophic Stock Implosions (CSI)** — permanent equity value destruction defined as an 80% drawdown from peak followed by a zombie period of at least 18 months without recovery above −20% from the crash low. The prediction task is:

$$\text{features}(t) \xrightarrow{\text{model}} \hat{y}(t+1) \tag{1.1}$$

where $\hat{y}(t+1) = 1$ if firm i enters a confirmed CSI event in year $t+1$. The one-year label shift is critical: features at year t predict the *following* year's outcome, preventing contemporaneous leakage from price features (e.g., `max_dd_12m`) into the CSI label triggered by the same crash.

1.1.1 Two Investor Use Cases

A central design decision is the distinction between two fundamentally different investor problems:

- **Ex-ante screening (M1):** At year t, *before* any significant drawdown is visible, screen firms based on fundamental accounting data alone. The model asks: *"Which firms should never enter the index?"* Features are limited to balance sheet, income statement, and cash flow ratios — no price information. This supports the **avoidance alpha** hypothesis: superior returns from systematically excluding structural losers before crashes occur.
- **In-crisis triage (M3):** At year t, a firm is already in drawdown. The model asks: *"Will this become a permanent implosion or recover?"* All features including drawdown severity (`max_dd_12m`) are available. This achieves higher AP but relies on contemporaneous crash signals unsuitable for ex-ante index construction.

The thesis prioritises M1 for index construction and uses M3 as a performance ceiling benchmark.

1.2 Four-Model Architecture

Four models are estimated to answer the two sub-questions of the thesis:

Table 1.1: Model Architecture Overview

Model	Feature Input	Classifier	Research Purpose
M1	Fundamentals only (∼452 features)	XGBoost + AutoGluon	Avoidance alpha; ex-ante screening
M2	VAE(fund) → $z_{1..24}$ + recon error	XGBoost + AutoGluon	Sub-Q 1: does VAE add signal?
M3	Full features (∼471 features)	XGBoost + AutoGluon	Performance ceiling; triage benchmark
M4	VAE(raw) → $z_{1..24}$ + recon error	XGBoost + AutoGluon	VAE compression of full features

- **M1 vs M2:** Answers Sub-Question 1 — does the VAE improve over raw fundamentals?
- **M1 vs M3:** Quantifies the *price-stripping cost* — how much predictive power is lost by removing price features for the purpose of ex-ante screening?
- **M3 vs M4:** Does VAE compression of the full feature set improve over raw features?

1.3 Label Definition

Following **?**, a CSI event for firm i at time t satisfies:

1. **Crash trigger:** Drawdown$(t) < C = -0.80$ from the rolling 3-year peak
2. **Zombie period:** Duration $T = 18$ months from crash low
3. **Non-recovery:** Cumulative return from crash low stays $< M = -0.20$ throughout the zombie window

A firm is right-censored ($y = \text{NA}$) if the zombie confirmation window extends beyond the sample end date (December 2024). Multiple CSI events per firm are permitted, which deviates from **?** and allows the model to learn from serial-offender dynamics. A 27-combination sensitivity grid ($C \times M \times T$) is used for robustness analysis in Chapter 12.

1.4 Temporal Validation Design

Standard k-fold cross-validation is inappropriate for time-series data due to look-ahead bias. An **expanding window** design is used throughout:

Table 1.2: Expanding Window CV Fold Structure

Fold	Training Period	Validation Period
1	*Burn-in* (no training data)	1993–1997
2	1993–2001	2002–2006
3	1993–2006	2007–2010
4	1993–2010	2011–2015
5	1993–2015	2016–2019 (= test set)

Note on Fold 1 (burn-in): The panel starts in 1993, but no observations exist prior to 1993 to serve as training data for Fold 1. The years 1993–1997 therefore function as a **burn-in period** — they populate the initial training window for Fold 2, ensuring that rolling 5-year features (e.g. `roll_mean_5y_earn_yld`) are populated with sufficient history before the first validation fold begins. Fold 1 is skipped for CV metric computation; all subsequent folds use 1993 as their training start date.
The three-way data split is:

- **Train** (1993–2015): hyperparameter optimisation via CV
- **Test** (2016–2019): model selection and threshold calibration
- **OOS** (2020–2024): final out-of-sample evaluation — never touched until evaluation

1.5 Evaluation Metrics

1.5.1 Primary Metric: Average Precision (AP / AUCPR)

Given class imbalance ($\approx 10\%$ CSI prevalence), AUC-ROC is misleading — a model achieving high ROC scores by correctly classifying the majority of non-CSI firms gains little practical value. **Average Precision** (area under the Precision-Recall curve) is the HPO objective and primary evaluation metric:

$$\text{AP} = \sum_n (R_n - R_{n-1}) \cdot P_n \tag{1.2}$$

Computed via `PRROC::pr.curve()` in R and `sklearn.metrics.average_precision_score` in Python to ensure consistency.

1.5.2 Secondary Metric: Recall at Fixed FPR

For index construction, the portfolio manager operates under a **false positive budget**: incorrectly excluding a high-growth "Ecstasy" stock (e.g., NVIDIA) is costly. **Recall at FPR** $\leq 3\%$ is the primary economic metric:

$$\text{R@FPR3} = \max\{\text{TPR}(\tau) : \text{FPR}(\tau) \leq 0.03\} \tag{1.3}$$

This measures: given that at most 3% of healthy firms are erroneously excluded, what fraction of true CSI events is captured?

1.5.3 Threshold Calibration

For the final index construction, threshold θ is calibrated on the test set to maximise F1-score (harmonic mean of precision and recall):

Table 1.3: Threshold Calibration — M3 XGBoost Test Set

FPR	Threshold	Recall	Precision	F1	Firms flagged
1%	0.844	24.0%	77.0%	36.6%	417
3%	0.789	47.3%	68.5%	55.9%	921
5%	**0.738**	**62.1%**	**63.1%**	**62.6%**	**1,313**
10%	0.612	80.9%	52.7%	63.8%	2,049

FPR = 5% is recommended for index construction (best F1 = 0.626).

Chapter 2

Data and Feature Engineering

2.1 Data Sources

1. **CRSP** (`crsp_a_stock.*`): Monthly equity prices, returns, market capitalisation. Universe: NYSE, AMEX, NASDAQ common equities with minimum 5-year listing lifetime. 19,647 unique `permno` identifiers.

2. **Compustat** (`comp.funda`): Annual fundamentals. Linked via CCM table (`crsp_a_ccm.*`) using LC/LU/LS link types with P/C primary marker. 176,495 rows, 18,526 unique firms.

3. **FRED API**: Macro variables — Federal Funds Rate, GDP growth, unemployment, 10-year Treasury yield, HY spread, VIX, CPI inflation, industrial production, recession indicator.

2.1.1 Look-Ahead Bias Mitigation

A **3-month reporting lag** is applied: Compustat fundamentals dated at fiscal year-end t are only joined to the annual panel for calendar year $t+1$ — reflecting the 10-K filing deadline of 90 days post fiscal year-end (`REPORTING_LAG_MONTHS = 3L`).

2.2 Feature Families

The full feature matrix contains ≈ 480 engineered features across 11 families:

Table 2.1: Feature Engineering Families

Family	Description	Count	Key Features
1	Point-in-time ratios	∼40	`earn_yld`, `roa`, `leverage`, Altman Z
2	YoY changes	∼41	`yoy_*`
3	Acceleration (2nd diff.)	∼41	`accel_*`
4	Expanding mean & vol	∼82	`expmean_*`, `expvol_*` (lagged by 1)
5	Peak deterioration	∼18	`peak_drop_*` (5-yr rolling peak)
6	Consecutive declines	∼10	`consec_decline_*`
7	Accounting momentum	∼15	`acct_mom_*`
8	Rolling 3-yr stats	∼90	mean, min, max, SD, trend, autocorr
9	Rolling 5-yr stats	∼90	same statistics over 5-yr window
10	Price momentum & vol	∼8	`mom_1m/3m/6m/24m`, `vol_12m/60m`, `max_dd_12m/60m`
11	Macro interactions	∼6	`interact_lev_rate`, `interact_ret_vix`, etc.

2.3 Critical Design Decisions in Feature Engineering

2.3.1 Rolling Peak for Drawdown (Family 10)

A key correction from a naive implementation: `max_dd_12m` and `max_dd_60m` use a **3-year rolling peak** rather than the all-time cumulative maximum.
Rationale: Using an all-time peak permanently depresses the drawdown reading for firms that experienced a distant historical high — a firm trading steadily at current levels appears to be in an artificially large drawdown relative to a decade-old peak. The rolling 3-year window measures *current deterioration*, not historical distance from a potentially irrelevant all-time high.

2.3.2 Multi-Year Price Feature Bug (Fixed)

Previous versions computed `vol_60m`, `mom_24m`, and `max_dd_60m` inside a `by=.(permno, year)` group containing at most 12 monthly rows. This caused:

- `vol_60m` $\equiv$ `vol_12m` (within-year SD only)
- `mom_24m` always NA (24 months of data never available within one calendar year)
- `max_dd_60m` always NA (same reason)

Fix: Two-step approach. Within-year features (`mom_1m/3m/6m`, `vol_12m`, `max_dd_12m`) computed inside the annual group. Multi-year features (`mom_24m`, `vol_60m`, `max_dd_60m`) computed via an explicit firm-level loop accumulating cumulative price history up to each year-end.

2.3.3 M1 Feature Set: Price Feature Stripping

For the ex-ante screening model (M1), all price-derived features are removed from the feature matrix. This produces `features_fund.rds` with ≈ 452 features. Stripped features include:

- All of Family 10: `mom_*`, `vol_*`, `max_dd_*`, `ann_return`, `log_return`
- All derivatives of price features (YoY, acceleration, expanding stats, rolling stats of `log_return` and `ann_return`)
- `interact_ret_vix` (price $\times$ macro interaction)

Note: `log_mkvalt` is **retained** as a firm size proxy (market capitalisation), not a price direction signal. Its derivatives are also retained.

2.3.4 Quantile Transform

All feature matrices are transformed to uniform $[0, 1]$ via QuantileTransformer **fitted on the training set only**, applied to test and OOS using the training ECDF. This prevents distribution shift between periods from affecting tree-based model split thresholds, and ensures neural networks within AutoGluon receive well-scaled inputs.

2.4 Panel Statistics

Table 2.2: Pipeline Output Sizes

Stage	Output File	Size
Universe	`universe.rds`	19,647 permno
Fundamentals	`fundamentals.rds`	176,495 rows, 18,526 permno
CSI Labels	`labels_base.rds`	9.66% prevalence, 16,466 events
Raw Panel	`panel_raw.rds`	127,649 rows, 12,895 permno, 90 cols
Full features	`features_raw.rds`	127,649 rows, ∼480 features
Fund features	`features_fund.rds`	127,649 rows, ∼452 features
Latent (fund VAE)	`features_latent_fund.parquet`	127,649 rows, 25 features
Latent (raw VAE)	`features_latent_raw.parquet`	127,649 rows, 25 features

Chapter 3

Variational Autoencoder

3.1 Architecture and Motivation

A **Beta-VAE** with an optional supervised classification head is implemented in Python (PyTorch) as `08B_Autoencoder.py`. The VAE serves two functions:

1. **Dimensionality reduction:** Compresses the ∼452 (fund) or ∼471 (raw) dimensional input vector x into a 24-dimensional latent representation z, capturing non-linear structural dependencies that linear PCA would miss.

2. **Anomaly detection feature:** The per-firm reconstruction error $\mathcal{L}_{\text{recon}} = ||x - \hat{x}||^2$ serves as an explicit feature in the downstream classifier. CSI firms that deviate from the "healthy manifold" produce higher reconstruction errors.

3.1.1 Architecture Details

Table 3.1: VAE Architecture

Component	**Specification**
Encoder	Input $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ $(z_\mu, z_{\log \sigma^2})$
Decoder	$z \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow$ output
Classifier head	$z_\mu \rightarrow 16 \rightarrow$ logit (optional, $\gamma = 0.1$)
Activation	LayerNorm + GELU
z dimension	24
Parameters (fund)	321,557
Parameters (raw)	331,304

3.2 Corrected ELBO Loss

The loss function is the corrected Evidence Lower BOund (ELBO):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}} + \gamma \cdot \mathcal{L}_{\text{clf}} \tag{3.1}$$

Key corrections vs a naive implementation:

1. **Loss reduction:** `reduction="none"` followed by `.sum(dim=1).mean()` — sums reconstruction error over the feature dimension first, then averages over the batch. This ensures $\mathcal{L}_{\text{recon}}$ and $\mathcal{L}_{\text{KL}}$ are on the same scale, preventing posterior collapse.

2. **PIT normalisation:** `output_distribution="normal"` (not uniform). Normal targets match the linear decoder's unbounded output range and align with Gaussian generative assumptions.

3. **Early stopping on validation loss:** 15% of training data held out for early stopping; best weights restored on validation loss minimum.

4. **Numerically stable classification loss:** Classifier head returns raw logits; `BCEWithLogitsLoss` applied (log-sum-exp trick).

3.3 Training Results

Both VAE runs (fund and raw inputs) completed 150 epochs without early stopping, with tight train/validation gaps indicating no overfitting.

Table 3.2: VAE Latent Space Diagnostics

Metric	**VAE-Fund (M2)**	**VAE-Raw (M4)**
Input features	452	471
Active dims (var $\geq 10^{-3}$)	**24 / 24**	**24 / 24**
Collapsed dims	0	0
Dim variance range	[0.83, 1.20]	[0.85, 1.19]
Recon error — CSI firms	0.4951	0.5124
Recon error — Non-CSI firms	0.3231	0.3320
Ratio (CSI / Non-CSI)	**1.532**	**1.543**
Test recon error p95	0.8874	0.9013

3.3.1 Key Finding: Anomaly Signal Confirmed

Both VAEs achieve a CSI/Non-CSI reconstruction error ratio > 1.5, confirming that the latent space has learned genuine anomaly structure. CSI firms are reconstructed $\approx 53\%$ worse than healthy firms, validating the Autoencoder hypothesis from Sub-Question 1. The ratio is marginally higher for the raw VAE (1.543 vs 1.532), consistent with price drawdown features providing additional distress signal.

3.3.2 What Worked Well

- **LayerNorm + GELU:** Superior to BatchNorm + ReLU for financial tabular data. LayerNorm normalises within each observation rather than across the batch, appropriate for high between-firm variance in financial ratios.

- **KL warmup (β ramp over 20 epochs):** Prevents the encoder from collapsing to the prior before the decoder has learned to reconstruct. Zero collapsed dimensions in both runs.

- **Gradient clipping (max_norm=1.0):** Stabilises training in early epochs when loss gradients are large.

- **Stratified train/val split:** Ensures class balance in the validation set used for early stopping.

- **Median imputation + winsorisation before PIT:** Converts Inf values to NaN, winsorises at 0.1%/99.9% of training distribution, then imputes with training medians. Prevents overflow in float32 tensors.

Chapter 4

Manual XGBoost Training

4.1 Overview

XGBoost models are trained via `09_Train.R` using Bayesian hyperparameter optimisation (HPO) with an expanding window CV objective. Two classifiers are trained for each feature set: the hand-tuned XGBoost here, and the AutoGluon ensemble in Chapter 5.

4.2 Bayesian HPO Design

- **Algorithm:** Bayesian Optimisation with Upper Confidence Bound (UCB, $\kappa = 2.576$) acquisition function via `rBayesianOptimization`
- **Iterations:** 10 random initialisation + 20 Bayesian = 30 total
- **Objective:** Maximise CV AUCPR (`eval_metric = "aucpr"`)
- **Class imbalance:** `scale_pos_weight = n_neg / n_pos` $\approx$ `8.94`
- **Nrounds:** 500 during HPO (early stopping at 20), 1000 during final model (early stopping at 50)

4.2.1 HPO Search Space

Table 4.1: Bayesian HPO Hyperparameter Bounds

Parameter	Lower Bound	Upper Bound
`eta` (learning rate)	0.01	0.30
`max_depth`	3	8
`subsample`	0.50	1.00
`colsample_bytree`	0.50	1.00
`min_child_weight`	1	20
`gamma`	0.00	5.00
`lambda` (L2)	0.00	5.00
`alpha` (L1)	0.00	5.00

4.3 Best HPO Configurations

4.3.1 M3 (Raw Features) — Optimal Parameters

- `eta = 0.019`, `max_depth = 5`, `subsample = 0.50`
- `colsample_bytree = 0.50`, `min_child_weight = 4`

- `gamma = 4.4`, `lambda = 5.0`, `alpha = 2.7`
- Optimal rounds: 110, CV AUCPR: **0.4855** (original run) / **0.5434** (revised 4-fold)

4.3.2 M1 (Fundamentals Only) — Optimal Parameters

- `eta = 0.010`, `max_depth = 3`, `subsample = 0.52`
- `colsample_bytree = 0.86`, `min_child_weight = 10`
- `gamma = 3.31`, `lambda` ≈ 0, `alpha = 1.39`
- Optimal rounds: 115, CV AUCPR: **0.4369**

The M1 optimal configuration is shallower (`max_depth = 3` vs 5 for M3) and more regularised, consistent with fewer informative features when price signals are removed.

4.4 XGBoost Results

Table 4.2: XGBoost Results — All Models

Model	CV AUCPR	Test AP	AUC	R@FPR3	R@FPR5	OOS AP
M3 XGB raw	0.5434	0.6823	0.9406	0.5131	0.6674	0.7653
M1 XGB fund	0.4369	0.5224	0.9001	0.2891	0.3693	0.6663
M2 XGB lat-fund	0.4131	0.5188	0.8877	0.3423	0.4637	0.6308
M4 XGB lat-raw	—	—	—	—	—	—
Paper benchmark	—	—	—	0.610	—	—

4.5 Feature Importance Analysis

4.5.1 M3 Top Features (after label shift + rolling peak fix)

Table 4.3: Top 10 Features by Gain — M3 XGBoost (Raw)

Rank	Feature	Gain	Interpretation
1	`max_dd_12m`	36.6%	Current drawdown severity (3-yr rolling peak)
2	`altman_z2`	14.0%	Retained earnings / assets
3	`ocf_per_share`	4.3%	Operating cash flow per share
4	`ann_return`	3.6%	Annual price return
5	`roll_mean_5y_earn_yld`	3.2%	5-yr avg earnings yield
6	`roll_min_3y_earn_yld`	2.3%	Worst earnings yield in 3-yr window

Note: Before the label shift correction, `max_dd_12m` dominated at 90.7% gain — near-tautological contemporaneous leakage. After the shift, gain concentration dropped to 36.6%, and the model uses a broad fundamental signal set.

4.5.2 M1 Top Features (Fundamentals Only)

Table 4.4: Top 10 Features by Gain — M1 XGBoost (Fund)

Rank	Feature	Gain	Interpretation
1	`altman_z2`	31.6%	Retained earnings / assets (solvency)
2	`roll_mean_5y_earn_yld`	10.8%	Persistent earnings quality
3	`roll_min_3y_earn_yld`	10.0%	Worst earnings in 3-yr window
4	`ocf_per_share`	9.9%	Cash generation capacity
5	`div_yield`	6.1%	Dividend cuts signal distress
6	`peak_drop_log_mkvalt`	5.6%	Size deterioration from 5-yr peak
7	`earn_yld`	3.9%	Current earnings yield
8	`roll_mean_3y_earn_yld`	3.4%	Rolling earnings quality
9	`roll_sd_5y_earn_yld`	2.2%	Earnings instability
10	`ebit_roa`	1.8%	EBIT return on assets

Without price features, the model relies on a combination of solvency (`altman_z2`: retained earnings depletion), earnings quality (`earn_yld` consistency and volatility), and cash generation capacity (`ocf_per_share`). This is economically coherent with the distress literature.

Chapter 5

AutoGluon Ensemble

5.1 Overview

AutoGluon (`09C_AutoGluon.py`) complements the hand-tuned XGBoost by automatically selecting and ensembling LightGBM, XGBoost, CatBoost, Random Forest, ExtraTrees, and Neural Networks. The implementation uses `good_quality` preset with a 3,600-second time limit per Stage 1 run.

5.2 Two-Stage Implementation

1. **Stage 1 — Full training:** AutoGluon trains on all training rows (1998–2010), using years 2011–2015 as a holdout for model selection. The best ensemble is retrained on train + holdout combined (`_FULL` suffix). Produces the deployable model.
2. **Stage 2 — Manual expanding window CV:** Three sequential AutoGluon fits (Folds 2/3/4, 900 seconds each, `medium_quality` preset) produce honest out-of-fold CV predictions for AP comparison with XGBoost.

5.3 Best Ensemble Composition

Table 5.1: AutoGluon Best Ensemble Weights by Model

Model	M1 (fund)	M3 (raw)	M4 (lat-raw)
LightGBMLarge	0.684	0.697 (best single)	—
CatBoost	0.158	—	—
RandomForestGini	0.105	—	—
LightGBMXT	0.053	—	0.421
NeuralNetTorch	—	—	0.421

LightGBM variants dominate in feature-rich settings (M1, M3); Neural Networks gain weight in the latent feature setting (M4) where the 25-dimensional input is well-suited to NN architectures.

5.4 Individual Model Hyperparameters and Performance

The `good_quality` preset trains 11 distinct model configurations. Their hyperparameters and holdout AP scores for M3 (raw features) are shown below — this is the most informative setting as it has the largest feature space and the clearest performance differentiation between model families.

Table 5.2: AutoGluon Individual Model Hyperparameters and Holdout AP — M3 (Raw Features)

Model	Holdout AP	Key Hyperparameters
LightGBMLarge	0.6972	lr=0.03, num_leaves=128, feature_fraction=0.9, min_data_in_leaf=3, rounds=4,687
LightGBMXT	0.6848	lr=0.05, extra_trees=True, rounds=1,855
LightGBM	0.6835	lr=0.05, rounds=720
CatBoost	0.6829	lr=0.05, iterations=768 (from 10,000 ceiling via early stopping)
RandomForestEntr	0.6799	n_estimators=300, max_depth=15, criterion=entropy
RandomForestGini	0.6728	n_estimators=300, max_depth=15, criterion=gini
ExtraTreesEntr	0.6586	n_estimators=300, max_depth=15, criterion=entropy
NeuralNetTorch	0.6529	lr=0.0003, 4 hidden layers (128 units), dropout=0.1, epochs=3 (early stop)
ExtraTreesGini	0.6527	n_estimators=300, max_depth=15, criterion=gini
NeuralNetFastAI	0.6470	lr=0.01, emb_drop=0.1, best epoch=5
XGBoost	0.6281	lr=0.10, n_estimators=109, default depth and subsampling
WeightedEnsemble	**0.6976**	Weights: LightGBMLarge=0.684, CatBoost=0.158, RF=0.105, LGBMXT=0.053

5.4.1 Why LightGBMLarge Dominates

LightGBMLarge achieves the highest single-model AP (0.697) and receives 68% of the ensemble weight in M1 and effectively 100% in M3. Three hyperparameter choices explain this:

1. **num_leaves=128 (vs default ∼31):** Financial distress follows highly non-linear, interaction-heavy patterns — e.g. high leverage is dangerous only when combined with low cash coverage and declining earnings. Deep trees with 128 leaves can learn these multi-way interactions. Shallower models miss the compound deterioration dynamics that precede CSI events.

2. **learning_rate=0.03 with 4,687 rounds:** A slow learning rate with many iterations allows the model to make incremental, precise corrections rather than overshooting. The 4,687 rounds (vs 720 for standard LightGBM) indicate the model continued finding marginal signal across a long training trajectory — consistent with the high-dimensional, noisy nature of fundamental accounting data where each additional boosting step extracts a small but genuine signal.

3. **feature_fraction=0.9:** Using 90% of features per tree introduces mild decorrelation between trees, reducing ensemble variance without the aggressive subsampling (50%) used in the hand-tuned XGBoost. With 452–471 features, 90% still provides full coverage of all feature families per split.

5.4.2 Why AutoGluon's XGBoost Underperforms

AutoGluon's internal XGBoost scores 0.628 on the M3 holdout — the weakest tree-based model — despite XGBoost being the strongest algorithm when hand-tuned (0.682 test AP). The reason is entirely the default hyperparameter configuration:

Table 5.3: AutoGluon XGBoost vs Hand-Tuned XGBoost: Parameter Comparison

Parameter	AutoGluon Default	Hand-Tuned (M3)	Effect of Default
learning_rate	0.10	0.019	5× too fast — overshoots optimal weights
subsample	1.00	0.50	No row subsampling — higher variance
colsample_bytree	1.00	0.50	All features used per split — higher variance
gamma	0.00	4.40	No split complexity penalty — memorises noise
lambda (L2)	1.00	5.00	Insufficient leaf weight regularisation
alpha (L1)	0.00	2.70	No sparsity — uses redundant features
n_rounds	109	110	Comparable (early stopping works correctly)

The Bayesian HPO explicitly searches for heavy regularisation ($\gamma = 4.4$, $\lambda = 5.0$, $\alpha = 2.7$) and aggressive subsampling (50%) to counteract overfitting on the noisy high-dimensional feature space. AutoGluon's default XGBoost configuration never explores this region, confirming the value of explicit hyperparameter search for this problem.

5.4.3 Why Neural Networks Underperform on Raw Features

Both NeuralNetTorch and NeuralNetFastAI score near the bottom of the M1/M3 leaderboard (AP 0.529–0.653) but gain significant ensemble weight in the latent settings (M4: 42% weight for NeuralNetTorch). Three structural reasons explain this pattern:

- **Early stopping fires aggressively:** NeuralNetTorch ran only 3 epochs on the M3 holdout before early stopping. With 452–471 features and heavy class imbalance (10% CSI prevalence), the network overfits within a handful of epochs and never reaches a well-generalised solution. The 4-layer, 128-unit architecture is not well-regularised for such high-dimensional tabular inputs.
- **Tabular financial data does not benefit from representation learning:** Financial ratios are already highly engineered — the model receives `roll_mean_5y_earn_yld` rather than raw financial statements. Neural networks excel at learning representations from raw, unstructured data. When features already encode domain-specific structure, the non-linear transformations of gradient boosting outperform neural architectures.
- **Improved performance on latent inputs (M4):** In the 25-dimensional latent space produced by the VAE, NeuralNetTorch scores competitively (0.555 holdout) and receives 42% ensemble weight. The VAE has already performed the representation learning step — the compact, dense, continuous latent space suits neural architectures better than the raw sparse tabular matrix.

5.4.4 Why Random Forests Are Competitive but Not Dominant

RandomForest and ExtraTrees score 0.651–0.680 on M3 — solid but consistently below LightGBM variants. Their fixed hyperparameters (`n_estimators=300`, `max_depth=15`) are not tuned by AutoGluon, and two structural limitations apply:

- **Bagging without boosting:** Random Forests reduce variance through averaging but cannot correct systematic bias. Boosted models iteratively correct residuals — better suited to the difficult minority-class detection task where early trees will systematically underpredict CSI probability.
- **No explicit class-imbalance handling:** AutoGluon's RF/XT implementations do not apply `scale_pos_weight` or class-balanced sampling. With 90% non-CSI prevalence, trees split predominantly on majority-class boundaries, reducing sensitivity on the CSI minority. The boosting algorithms handle this implicitly through residual weighting.

5.5 AutoGluon Results

Table 5.4: AutoGluon Results — All Models (Stage 1)

Model	CV AP	Test AP	AUC	R@FPR3	R@FPR5	OOS AP
M3 AG raw	0.6656	0.7576	0.9601	0.6397	0.7670	0.8513
M1 AG fund	0.5809	0.6546	0.9337	0.4936	0.6337	0.8007
M4 AG lat-raw	0.5199	0.5897	0.9120	0.4090	0.5431	0.7194
M2 AG lat-fund	0.4854	0.5513	0.8940	0.3693	0.4951	0.6853
Paper benchmark	—	—	—	0.610	—	—

5.5.1 OOS Year-Level Breakdown (M3 AutoGluon)

Table 5.5: OOS Year-Level Results — M3 AutoGluon

Year	Prevalence	AP	R@FPR3	R@FPR5
2016	12.8%	0.713	0.550	0.655
2017	16.1%	0.810	0.624	0.742
2018	10.2%	0.702	0.563	0.664
2019	9.2%	0.577	0.461	0.602
2020	16.0%	0.792	0.596	0.696
2021	16.5%	0.786	0.588	0.662
2022	**9.8%**	**0.863**	**0.768**	**0.814**
2023	—	*excluded*	—	—

2023 excluded: right-censoring (zombie confirmation window not closed).

2022 strongest year: rate-hiking cycle activates leverage/interest coverage features.

5.6 AutoGluon Feature Importance

5.6.1 M1 (Fundamentals Only) — Key Finding

Table 5.6: Top 10 Features by Permutation Importance — M1 AutoGluon (Fund)

Feature	Importance	Interpretation
`div_yield`	0.00130	Dividend cuts signal funding stress
`acct_mom_log_mkvalt`	0.00068	Recent size momentum (accounting-based)
`fedfunds`	0.00055	Macro regime — rate environment
`expvol_log_mkvalt`	0.00048	Long-run size volatility
`roll_sd_5y_earn_yld`	0.00047	Earnings yield instability
`expmean_div_yield`	0.00034	Long-run dividend baseline
`roll_min_3y_roic`	0.00025	Worst return on invested capital
`roll_min_5y_earn_yld`	0.00024	Worst earnings yield in 5-yr window
`expvol_ocf_margin`	0.00022	Cash flow margin volatility
`altman_z2`	0.00021	Retained earnings / assets

The M1 feature importance tells a coherent fundamental distress story: *dividend sustainability*, *earnings quality consistency*, *cash flow stability*, and *solvency* (retained earnings depletion) are the leading predictors of future CSI when no price information is available.

Chapter 6

Overall Summary and Interpretation

6.1 Complete Results Matrix

Table 6.1: Complete M1–M4 Results: XGBoost and AutoGluon

Model	Algo	CV AP	Test AP	AUC	R@FPR3	R@FPR5	OOS AP
M3 raw	AutoGluon	0.6656	0.7576	0.9601	0.6397	0.7670	0.8513
M3 raw	XGBoost	0.5434	0.6823	0.9406	0.5131	0.6674	0.7653
M1 fund	AutoGluon	0.5809	0.6546	0.9337	0.4936	0.6337	0.8007
M1 fund	XGBoost	0.4369	0.5224	0.9001	0.2891	0.3693	0.6663
M4 lat-raw	AutoGluon	0.5199	0.5897	0.9120	0.4090	0.5431	0.7194
M2 lat-fund	AutoGluon	0.4854	0.5513	0.8940	0.3693	0.4951	0.6853
M2 lat-fund	XGBoost	0.4131	0.5188	0.8877	0.3423	0.4637	0.6308
Paper benchmark (**?**)		—	—	—	**0.610**	—	—

6.2 Five Key Findings

6.2.1 Finding 1: M3 Beats the Paper Benchmark

AutoGluon M3 achieves R@FPR3 = 0.640 on the test set and 0.721 on OOS, surpassing the **?** benchmark of 0.610 by +3pp on test and +11pp on OOS. The OOS period (2020–2022) includes COVID, a rate-hiking cycle, and a subsequent correction — genuine macro stress that the model handles well. 2022 is the strongest individual year (AP = 0.863), consistent with the rate-hiking cycle activating leverage and interest coverage features that were dormant during the near-zero rate environment.

6.2.2 Finding 2: M1 Validates the Avoidance Alpha Hypothesis

M1 (AutoGluon, fundamentals only) achieves Test AP = 0.655 and OOS AP = 0.801, beating the paper benchmark on OOS **without any price features**. This directly validates the avoidance alpha hypothesis: accounting data alone is sufficient to predict future CSI events one year ahead, before crash signals are visible in equity prices. The index construction rationale is thus defensible.

6.2.3 Finding 3: AutoGluon Dominates XGBoost Across All Models

Table 6.2: AutoGluon vs XGBoost — Test AP Improvement

Model	XGB Test AP	AG Test AP	Δ
M3 (raw)	0.6823	0.7576	**+0.075**
M1 (fund)	0.5224	0.6546	**+0.132**
M2 (lat-fund)	0.5188	0.5513	**+0.033**

The AutoGluon advantage is largest for M1 (+0.132), where LightGBM's ensemble captures non-linear interactions between fundamental ratios more effectively than a single hand-tuned XGBoost tree.

6.2.4 Finding 4: VAE Compression Hurts Predictive Performance (Sub-Q 1)

Table 6.3: VAE Compression Cost: M1 vs M2, M3 vs M4

Comparison	Algo	Δ CV AP	Δ Test AP	Δ R@FPR3
M1 → M2 (fund: raw vs VAE)	AutoGluon	−0.095	−0.103	−0.124
M1 → M2 (fund: raw vs VAE)	XGBoost	−0.024	−0.004	+0.053
M3 → M4 (raw: raw vs VAE)	AutoGluon	−0.146	−0.168	−0.231

The pattern is consistent across classifiers and input types: **VAE compression loses more discriminative signal than it creates**. Compressing 452 fundamental features into 25 latent dimensions discards approximately 10pp of test AP. The VAE's anomaly detection capability (recon error ratio > 1.5) adds genuine signal — `vae_recon_error` ranks 3rd by importance in M2 — but cannot compensate for the information loss from the bottleneck.

6.2.5 Finding 5: The CV↔Test Gap is Structural, Not Leakage

Table 6.4: CV↔Test Gap Analysis

Model	CV AP	Test AP	Gap
AG M3 raw	0.666	0.758	+0.092
AG M1 fund	0.581	0.655	+0.074
AG M2 lat-fund	0.485	0.551	+0.066
AG M4 lat-raw	0.520	0.590	+0.070

The gap is positive (test > CV) for all models and narrows as models simplify. This is consistent with early CV folds (1998–2006) being genuinely harder — fewer training observations, dotcom/pre-GFC macro regime — rather than data leakage. The benign 2016–2019 test period (low macro stress) makes CSI prediction *easier* than earlier decades.

6.3 Thesis Narrative

M3 (full features, AutoGluon ensemble) achieves state-of-the-art AP, beating the **?** *benchmark by +11pp on OOS R@FPR3. However, M3 relies on contemporaneous drawdown signals (`max_dd_12m`: 36.6% XGBoost gain) that are only observable after a crash has already begun — unsuitable for ex-ante index construction. M1 (fundamentals only) removes all price features and demonstrates that accounting data alone is sufficient to beat the paper benchmark on OOS, validating the avoidance alpha hypothesis. The VAE (M2, M4) successfully learns anomaly structure — reconstruction*

error ratios of 1.53–1.54 *confirm that CSI firms deviate from the healthy manifold — but the 24-dimensional bottleneck discards more discriminative information than it creates. AutoGluon's LightGBM ensemble outperforms hand-tuned XGBoost across all models, with the largest advantage in the fundamentals-only setting (+13.2pp test AP). The recommended model for index construction is* ***M1 AutoGluon*** *at FPR* $= 5\%$ *(*$\theta = 0.738$*, F1* $= 0.626$*): it screens firms before crashes are visible, achieves R@FPR3 of* 0.494 *on test and* 0.658 *on OOS, and relies entirely on accounting information available 3 months after fiscal year-end.*

6.4 Open Items and Next Steps

1. **11_Results.R**: Index construction and backtest — apply M1 AG predictions at $\theta = 0.738$ to construct the crash-filtered index, backtest vs Russell 3000 and MSCI Min-Vol benchmark.

2. **12_Robustness.R**: 27-combination CSI parameter grid ($C \times M \times T$) — verify that M1 findings are stable across alternative CSI definitions.

3. **XGB M4**: Run `09_Train.R` with `MODELS_TO_RUN <- c("latent_raw")` to complete the XGBoost M4 result for the full comparison table.

4. **Logistic Regression baseline**: Sub-Question 2 requires comparison against logistic regression and inverted Altman Z-Score as linear baselines.

5. **Look-Ahead Bias section**: Chapter 6.3 placeholder (`TBD`) needs to be completed documenting the 3-month reporting lag implementation in `06_Merge.R`.

Bibliography